RAJADHANI INSTITUTE OF ENGINEERING AND TECHNOLOGY
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MEDBUDDY: AI-Powered Medical Chatbot with Retrieval-Augmented Generation

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Abstract

- AI tool aimed at revolutionizing medical education and managing student stress.
- Utilizes Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG).
- Ensures strict groundedness, completely eliminating AI hallucinations.
- Integrates Streamlit, LlamaIndex, and TruLens for a seamless user experience.

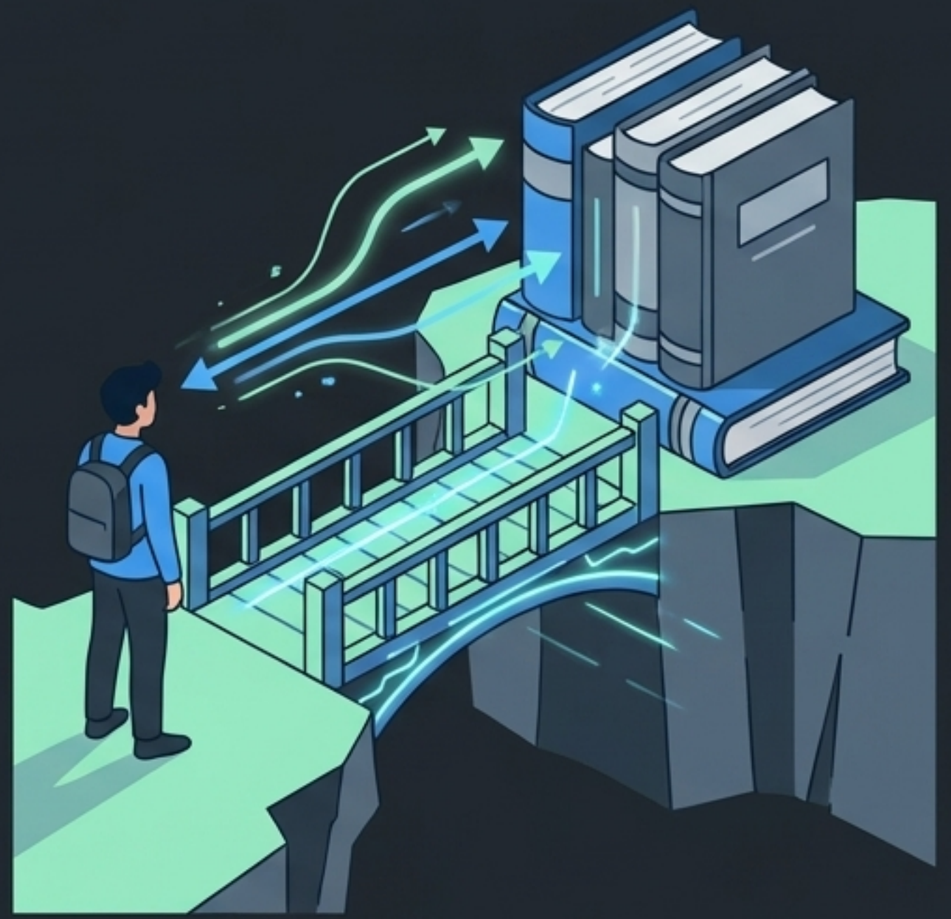
84%

Response accuracy rate for single medical textbooks.

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Introduction – Overview

- Rising demand for skilled professionals necessitates better medical preparation tools.
- Med-Buddy serves as a friendly digital assistant tailored to the user's focus area.
- Generic AI models lack transparency and carry a high risk of hallucination.
- The application shifts focus from slow data retrieval to true medical comprehension.



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Introduction – Motivation

Democratizes safe access to medical knowledge by isolating it from generative AI creativity.

Embeddings

Turning human sentences into mathematical number-lists to capture exact context.

Vector Database

Specialized storage for instant matching of user questions to verified text.

LLM Grounding

Strict rules forcing the AI to generate answers using only uploaded documents.

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Libraries Used

Streamlit

Powers the entire web-based interface and session management.

LlamaIndex & FastEmbed

Core data framework and local, efficient text embeddings.

Google Generative AI & Groq

Base Gemini LLM and high-performance LLM provider.

TruLens

Performs real-time evaluation of AI groundedness and answer relevance.

PyMuPDFReader & BeautifulSoup4

PDF and structural data extraction.

LiteLLM, NumPy, Hashlib, OS

System, mathematical, and execution management.

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Literature Survey – Overview

Examines the balance between technical innovation and practical healthcare benefits.

Explores the transformative role of AI chatbots in medical learning and public health.

Highlights how Large Language Models fundamentally alter knowledge delivery.

Emphasizes the crucial need for responsible AI integration and rigorous accuracy validation.

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Literature Survey – AI Chatbots in Medical Education and Public Health

Accelerated Potential

- Investigates standard LLMs in medical learning and vaccination training.
- Demonstrates that AI tools effectively accelerate medical research cycles.

Critical Barriers

- Uncovers a high risk of authenticity issues without a dedicated retrieval system.
- Concludes that professional verification and reliable source-tracking are mandatory.

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Literature Survey – RAG in Medical Chatbots

- Addresses the critical risks of general AI lacking domain-specific factual accuracy.
- Investigates systems that use AI to assess candidate responses and medical queries.
- Proves that grounding LLM capabilities in secure text eliminates outside fabrication.
- Highlights the shift toward systems that provide transparent, citation-backed answers.



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Literature Survey – Embedding Models and Semantic Search

- Analyzes the process of converting complex medical text into mathematical vectors.
- Reviews QA systems using pre-trained BERT models and semantic reasoning.
- Evaluates multiple embedding models to optimize retrieval performance (e.g., E5, BGE M3).
- Identifies the need for local embeddings to bypass internet bottlenecks and cloud costs.



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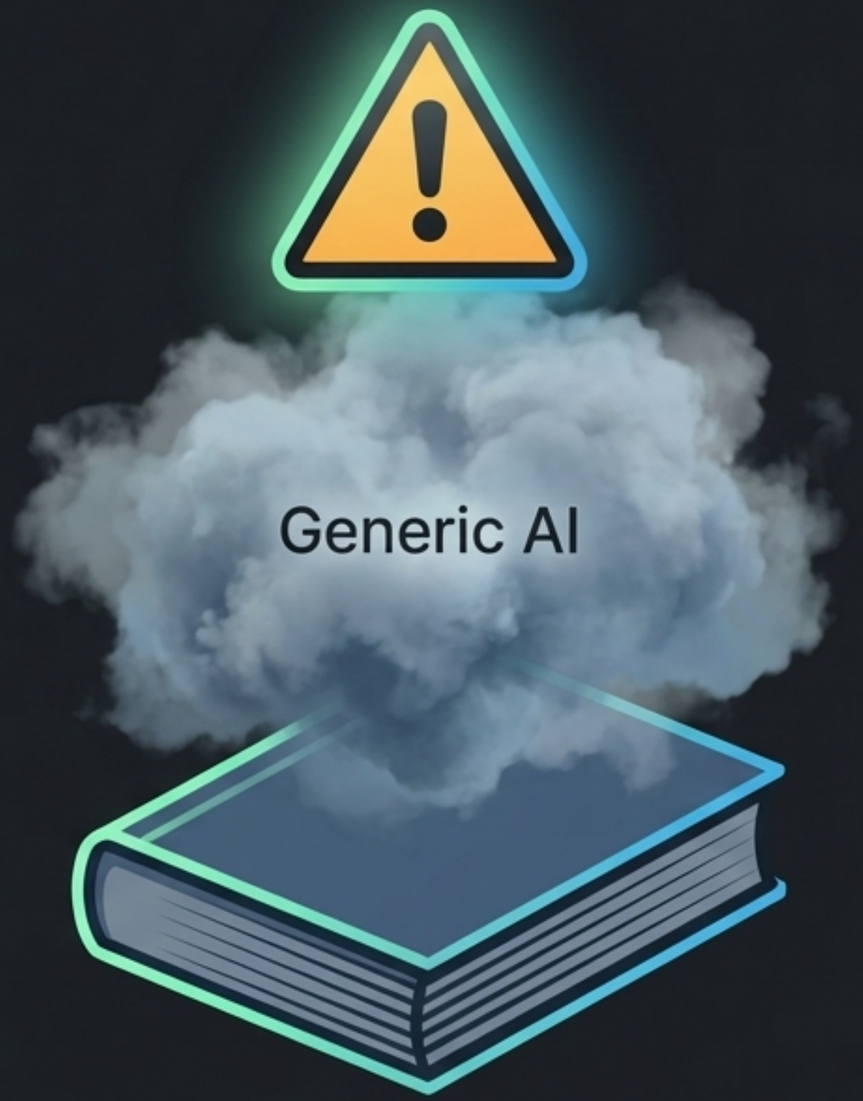
Literature Survey – LLMs in Healthcare

- Evaluates foundational models for high-level reasoning and context synthesis.
- Exposes massive security vulnerabilities when proprietary data is sent to external clouds.
- Highlights how purely cloud-based systems violate strict patient confidentiality guidelines.
- Stresses the necessity of decoupling data ingestion from cloud LLM generation.

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Problem Identification

- **Lack of Transparency:** Generic AI is not concrete about the source of its information.
- **The Hallucination Risk:** Standard LLMs have a high possibility of fabricating facts.
- **Vulnerability:** Students are exposed to false information or data unrelated to their syllabus.
- **Critical Consequences:** Misleading medical students with wrong information might cost lives.



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Existing System



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Disadvantages of Existing System

Severe Data Privacy Risks

Uploading medical documents to third-party servers violates privacy laws.

Exorbitant API Costs

Indexing large textbook volumes consumes millions of tokens.

High Latency

Bottlenecked by internet bandwidth, making large document vectorization painfully slow.

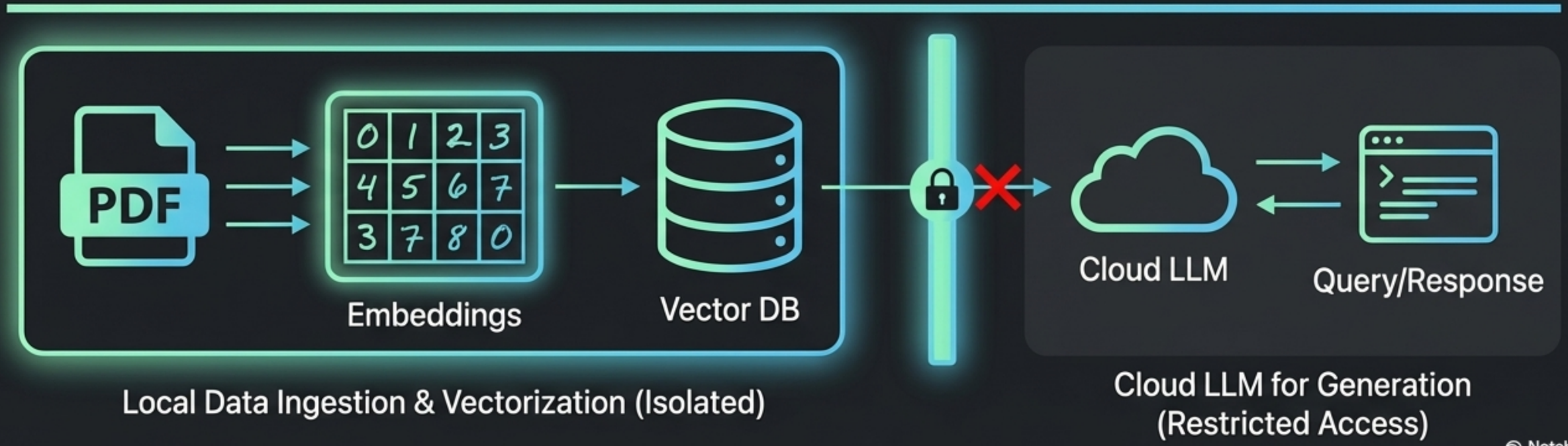
Rate Limiting

API crashes and timeouts make the system fragile for bulk data processing.

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Proposed System – MedBuddy Hybrid Framework

- Introduces a sophisticated, hybrid RAG framework tailored for healthcare environments.
- Bifurcates the workload to maximize both security and artificial intelligence capabilities.
- Processes, cleans, chunks, and vectorizes all medical data locally on the host machine.
- Explicitly restricts the cloud-based LLM from accessing the raw data ingestion phase.



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Our Contribution – Local Embedding Integration

- Integrates FastEmbed for local, efficient text embeddings, cutting external cloud API reliance.
- Keeps the vector database (FAISS/ChromaDB) entirely on the secure local file system.
- Slashes operational costs by reducing upload-phase token usage to zero.
- Overcomes rate limits and API exhaustion while maintaining an exceptionally high factual accuracy.



Zero Upload Token Cost

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Software and Hardware Requirements

Software

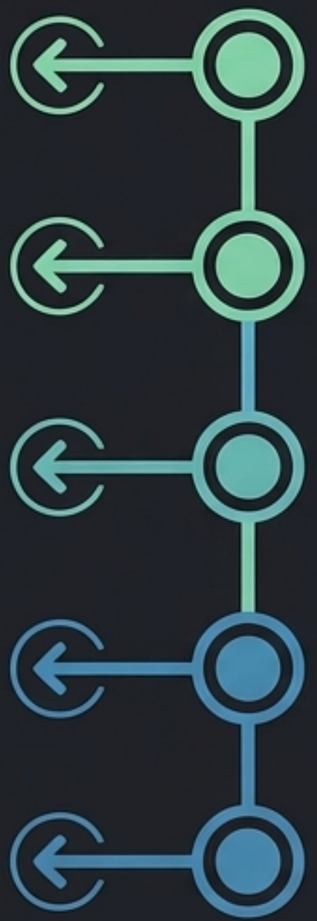
- Python (Core)
- LlamaIndex
- TruLens
- Streamlit
- Google Gemini / Groq APIs

Hardware

- **Processor:** Intel i5 or i7, AMD Ryzen 5 or 7
- **Memory:** Minimum 8GB RAM required for local data processing
- **Storage:** SSD (solid state drive) recommended, minimum 256GB
- **Operating System:** 64-bit version of Windows 10/11, macOS, or Linux

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Project Plan



Jan 16, 2026: Initial initiation and UI set-up.

Feb 1, 2026: Building Streamlit interface and configuring Python libraries.

Feb 15, 2026: Data processing module implementing PDF chunking and embedding.

Mar 1, 2026: Integrating LlamaIndex vector database and APIs.

Mar 6, 2026: 50% project completion for first evaluation.

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Base Paper Implementation – Placeholder

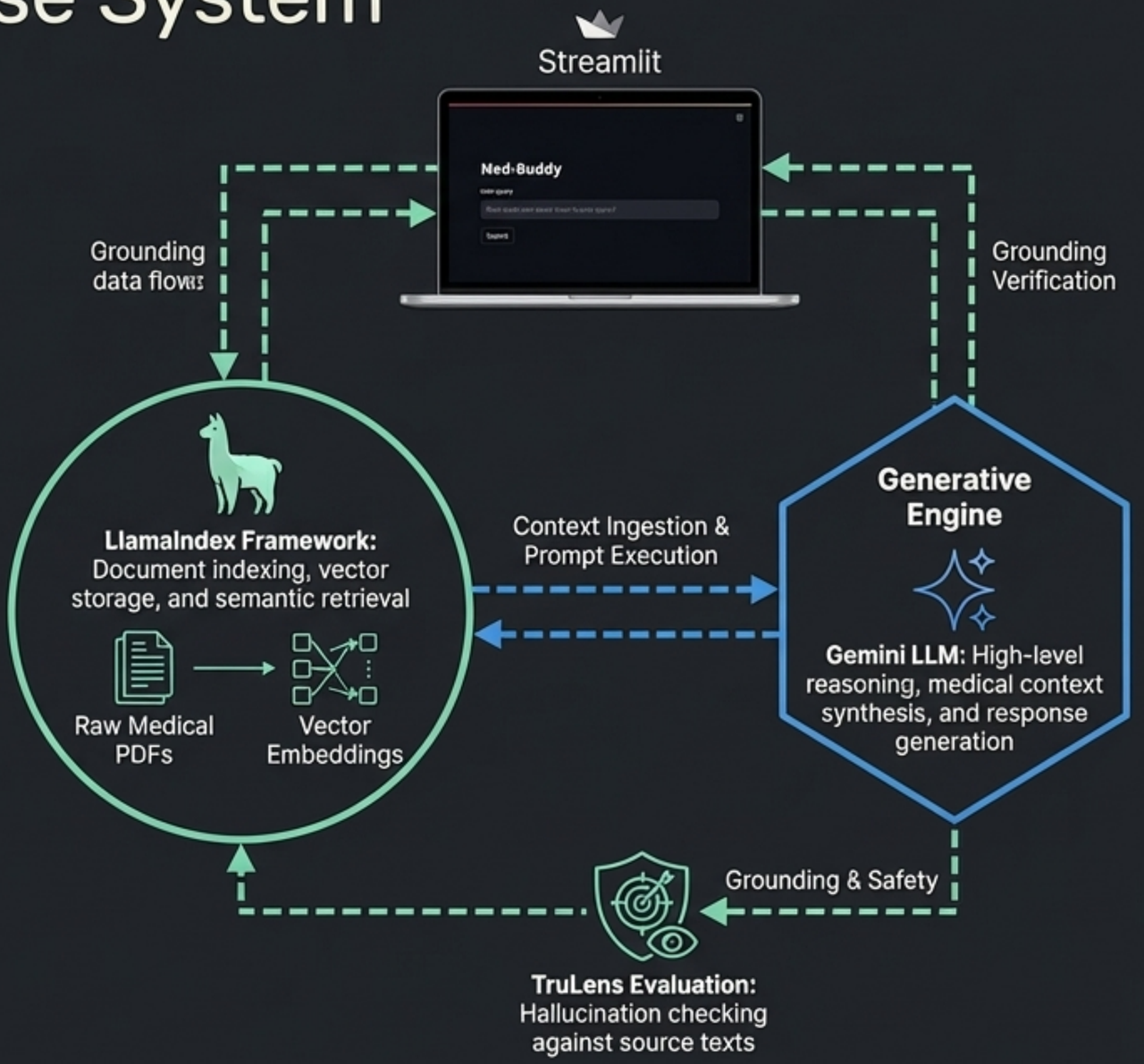
Placeholder for diagram: A flowchart illustrating the old monolithic cloud architecture where both document embedding and user queries are sent directly to the external Gemini API over the internet.

Labels to include: Raw Medical PDFs, Internet Transmission, Gemini Embedding Endpoint, Cloud Vector DB, Gemini Pro Generative Model, High API Cost / Latency.

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System Architecture – Base System

- Uses Streamlit for seamless, real-time query input and response rendering.
- Leverages the LlamaIndex Framework for document indexing and semantic retrieval.
- Converts raw medical PDFs to vector embeddings via cloud services.
- Employs Gemini LLM for high-level reasoning and medical context synthesis.
- Integrates TruLens Evaluation for hallucination checking against source texts.



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Our Contribution – Placeholder

Placeholder for diagram: A side-by-side split visual contrasting the old cloud ingestion method with the new local embedding method, highlighting data staying securely on the host machine.

Labels to include: Cloud-Only Ingestion (Vulnerable), Local FastEmbed Processing (Secure), Zero Token Cost, Protected File System, API Independence.

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